Special/uncommon versions of the Poisson

The Poisson distribution is

$$\text{Prob}(y) = \frac{\lambda^y}{y!}\exp(-\lambda)$$

for responses $y = 0, 1, 2, \ldots$, where

λ**:** the expected value.

Various special versions can be defined from here.

Special1

Here we consider a 1-inflated modification where there its known there are no zeros

$$\text{Prob}(y) = p \times 1_{[y=1]} + \frac{1-p}{1-\exp(-\lambda)} \times \frac{\lambda^y}{y!}\exp(-\lambda), \qquad y = 1, 2, \ldots$$

Link-function

λ is linked to the linear predictor by

$$\lambda = E\exp(\eta)$$

where $E > 0$ is a known constant (or $\log(E)$ is the offset of η).

Hyperparameters

Special1 have hyperparmeter p which is represented as

$$p = \frac{\exp(\theta)}{1+\exp(\theta)}$$

and the prior is given on θ

doc `The Poisson.special1 likelihood`
hyper
  theta
    hyperid `56100`
    name `logit probability`
    short.name `prob`
    output.name `one-probability parameter for poisson.special1`
    output.name.intern `intern one-probability parameter for poisson.special1`
    initial `-1`
    fixed `FALSE`
    prior `gaussian`
    param `-1 0.2`
    to.theta `function(x) log(x / (1 - x))`
    from.theta `function(x) exp(x) / (1 + exp(x))`
survival `FALSE`
discrete `TRUE`
link `default log`
pdf `poisson-special`

Specification

- `family="poisson.special1"`
- Required arguments: (integer-valued) y and E

Example

In the following example we estimate the parameters in a simulated example with Poisson responses.

```
n <- 300
a <- 1
b <- 1
p <- 0.2
x <- rnorm(n, sd = 0.2)
mu <- exp(a+b*x)
y.max <- ceiling(max(mu + 10*sqrt(mu)))
y <- numeric(n)

for(i in 1:n) {
    yy <- 1:y.max
    dy <- dpois(yy, lambda = mu[i])
    dy <- dy/sum(dy)
    dy <- dy * (1-p)
    dy[1] <- dy[1] + p
    y[i] <- sample(x = yy, size = 1, prob = dy)
}

r <- inla(y ~ 1 + x,
          data = data.frame(y, x),
          family = "poisson.special1")
summary(r)
```

Notes